

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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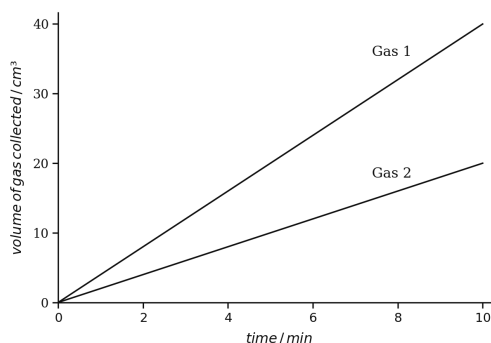
Chemistry

- 1 A constant direct electric current is passed through a dilute aqueous solution of sodium sulfate, $\text{Na}_2\text{SO}_4(\text{aq})$, using inert platinum electrodes. Gases are evolved at both electrodes.

The graph shows the volume of each gas collected over time at constant temperature and pressure.

Which of the following statements is/are correct?

1. Gas 1 is produced at the cathode (negative electrode).
2. In the external circuit, electrons flow away from the electrode producing Gas 2 towards the electrode producing Gas 1.
3. As the electrolysis proceeds, the concentration of Na_2SO_4 in the solution decreases.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 2 A metabolic pathway converts one molecule of substrate S into two molecules of product P ($1S \rightarrow 2P$). The overall process has a **net** yield of 4 molecules of ATP per molecule of S consumed.

The pathway proceeds in two sequential stages:

Stage	Event	ATP Change (per molecule of S)
1	Activation of S	Consumes 2 ATP
2	Conversion to P	Produces ATP

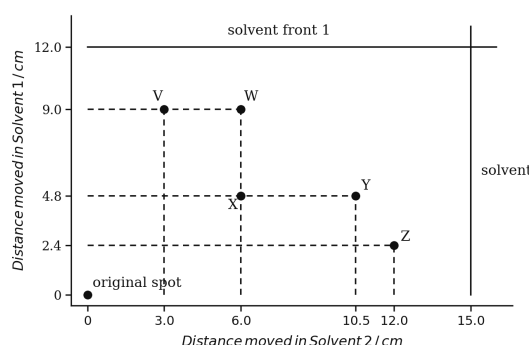
How many molecules of ATP are produced in Stage 2 for **each** molecule of P formed?

- A 1
B 2
C 3
D 4
E 6
- 3 A student carried out a two-dimensional thin-layer chromatography experiment to separate a mixture containing five organic compounds: V, W, X, Y, and Z.

A spot of the mixture was applied at the origin (0, 0). The plate was developed in Solvent 1 until the solvent front reached 12.0 cm along the vertical axis. The plate was dried, rotated by 90° , and developed in Solvent 2 until the solvent front reached 15.0 cm along the horizontal axis.

The final positions of the five compounds are shown on the chromatogram below.

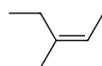
Which of the following statements is correct?



- A Developing the mixture using Solvent 1 alone would resolve all five compounds into separate spots.
B Developing the mixture using Solvent 2 alone would resolve all five compounds into separate spots.
C Compound X has the same R_f value in Solvent 1 as it has in Solvent 2.
D Compound Z has a greater R_f value in Solvent 1 than in Solvent 2.
E The R_f value of compound Y is 0.70 in Solvent 1 and 0.40 in Solvent 2.

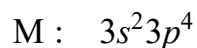
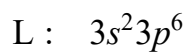
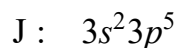
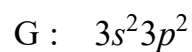
- 4 The alkene shown below reacts with hydrogen bromide, HBr , in an addition reaction:
Which of the following products is/are formed in this reaction?

- 1 $\text{CH}_3\text{CH}_2\text{CBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$
- 2 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHBrCH}_3$
- 3 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{Br}$



- A none of them
 - B 1 only
 - C 2 only
 - D 3 only
 - E 1 and 2 only
 - F 1 and 3 only
 - G 2 and 3 only
 - H 1, 2 and 3
- 5 Element X has atomic number 4. It reacts with chlorine to form a binary compound.
Which of the following options correctly identifies the formula of this compound and describes its structure in the solid state?
- A XCl_2 , ionic lattice
 - B XCl_2 , simple molecular
 - C XCl_2 , polymeric chains
 - D XCl_4 , simple molecular
 - E XCl , ionic lattice

- 6 The ground state electronic configurations of the outer shells of four elements, denoted by the letters G, J, L, and M, are given below:



Which of the following correctly lists these elements in order of **increasing** chemical reactivity?

- A L, G, M, J
 - B G, M, J, L
 - C L, J, M, G
 - D J, M, G, L
- 7 Concentrated aqueous solutions of the five ionic compounds listed below are electrolysed using inert carbon electrodes.

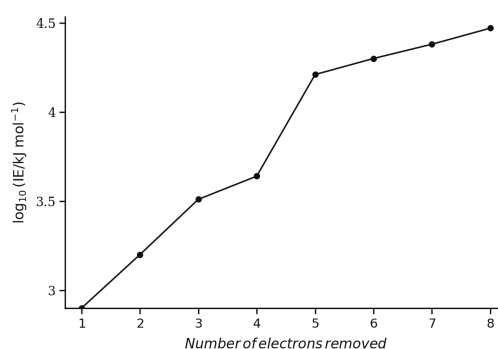
Which compound produces the elemental metal at the cathode and the elemental halogen at the anode?

- A NaCl
- B MgBr₂
- C AgF
- D CuCl₂
- E KI

- 8 The graph shows the first eight successive ionisation energies (IE) of an element Q , plotted as $\log_{10}(\text{IE}/\text{kJ mol}^{-1})$ against the number of electrons removed. Element Q belongs to Period 3 of the Periodic Table.

Which row in the table correctly identifies the Group of element Q , the formula of its stable chloride, and the total number of unpaired electrons in a ground-state neutral atom of Q ?

	Group	Formula of chloride	Number of unpaired electrons
A	13	QCl_3	1
B	13	QCl_3	3
C	14	QCl_4	0
D	14	QCl_4	2
E	14	QCl_4	4
F	15	QCl_5	1
G	15	QCl_5	3
H	15	QCl_3	5



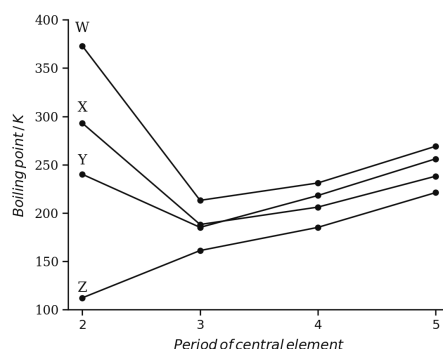
- 9 The graph shows the boiling points of the hydrides of elements in Groups 14, 15, 16, and 17 for Periods 2 to 5.

Which of the following statements is/are correct?

1 Curve W represents the hydrides of Group 16, and the anomalously high boiling point of its Period 2 hydride is due to hydrogen bonding.

2 From Period 3 to Period 5 along each curve, the boiling point increases because the intramolecular covalent bonds become progressively stronger.

3 Curve Z represents the hydrides of Group 14, which show no anomaly in Period 2 because CH_4 molecules cannot form hydrogen bonds.



- A none of them
 B 1 only
 C 2 only
 D 3 only
 E 1 and 2 only
 F 1 and 3 only
 G 2 and 3 only
 H 1, 2 and 3

- 10** An extract from a plant secretion contains three volatile liquids: P, Q, and R. A student attempts to purify each liquid using the following procedure:

1. Place the mixture in a separating funnel and allow the layers to settle.
2. Run the layers into separate flasks.
3. Perform fractional distillation on the contents of each flask.

The properties of the pure liquids are shown below:

Liquid	Density / g cm ⁻³	Boiling point / °C	Miscibility with P
P	0.79	78	-
Q	0.81	78	Miscible
R	1.00	78	Immiscible

Which of the liquids can be isolated in a pure form?

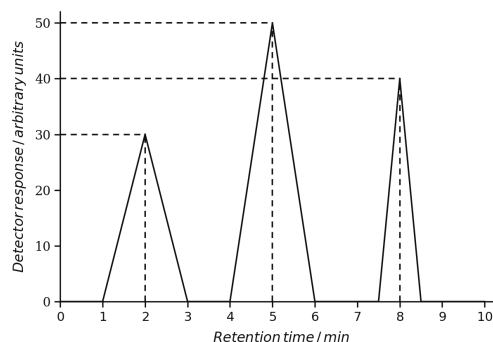
- A** P only
- B** R only
- C** P and Q only
- D** P and R only
- E** Q and R only
- F** P, Q and R
- G** None of them

- 11** A sample containing a mixture of three volatile organic compounds is analysed using gas chromatography with a non-polar stationary phase and an unreactive carrier gas as the mobile phase.

The detector response is directly proportional to the concentration of each compound, and each chromatographic peak can be approximated as a triangle (area = $\frac{1}{2} \times \text{base} \times \text{height}$).

The resulting chromatogram is shown below.

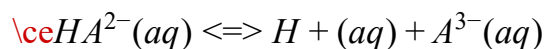
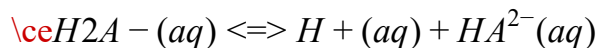
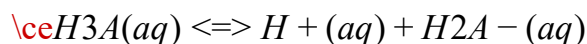
What percentage of the total peak area corresponds to the compound with the strongest attraction to the stationary phase?



- A** 20%
- B** 25%
- C** 30%
- D** 33%
- E** 40%
- F** 50%

- 12 Citric acid, whose structure is shown below, is a naturally occurring organic acid found in citrus fruits.

In aqueous solution, citric acid is a weak acid that ionises in three successive equilibrium steps:

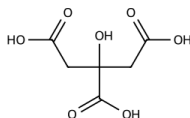


where H_3A represents citric acid.

A 0.10 mol dm^{-3} aqueous solution of hydrochloric acid, $\text{HCl}(\text{aq})$, has a pH of 1.0 at 25°C .

Which of the following statements about these solutions is/are correct?

- 1 The pH of a 0.10 mol dm^{-3} aqueous solution of citric acid at 25°C is greater than 1.0.
- 2 In aqueous solution, the dihydrogen citrate ion, H_2A^- , can act as both a Brønsted–Lowry acid and a Brønsted–Lowry base.
- 3 Exactly 30.0 cm^3 of 0.10 mol dm^{-3} barium hydroxide solution, $\text{Ba}(\text{OH})_2(\text{aq})$, is required to completely neutralise 20.0 cm^3 of 0.10 mol dm^{-3} citric acid solution.

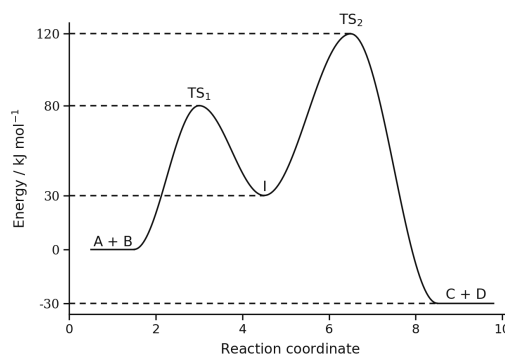


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 13 The energy profile for a two-step reaction $A + B \rightarrow C + D$ proceeding via an intermediate I is shown in the diagram.

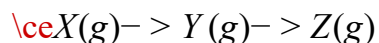
Which of the following statements is/are correct?

- 1 The overall reaction is exothermic with an enthalpy change of $\Delta H = -30 \text{ kJ mol}^{-1}$.
- 2 The activation energy for the rate-determining step of the forward reaction is $+90 \text{ kJ mol}^{-1}$.
- 3 The activation energy for the reverse reaction ($C + D \rightarrow A + B$) is $+120 \text{ kJ mol}^{-1}$.



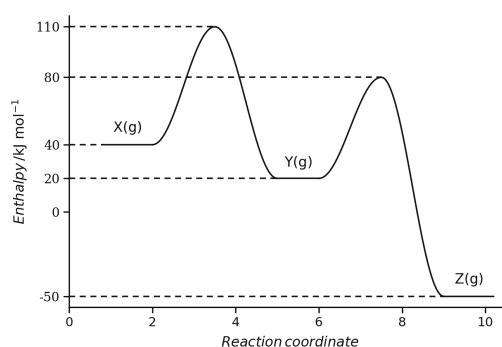
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 14 The diagram shows the enthalpy profile for a gaseous reaction that proceeds via a two-step mechanism:



Which of the following statements is/are correct?

- 1 The first step, $X(g) \rightarrow Y(g)$, is the rate-determining step of the forward reaction.
- 2 The activation energy for the reverse reaction of the first step, $Y(g) \rightarrow X(g)$, is $+70 \text{ kJ mol}^{-1}$.
- 3 The overall enthalpy change for the reaction $X(g) \rightarrow Z(g)$ is -90 kJ mol^{-1} .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3
- 15 Element X has the atomic number 15. Which of the following correctly describes the formula and the principal type of bonding in the simplest hydride of X?
- A Formula: XH_2 ; Bonding: covalent
- B Formula: XH_3 ; Bonding: ionic
- C Formula: XH_3 ; Bonding: covalent
- D Formula: XH_4 ; Bonding: covalent
- E Formula: XH_5 ; Bonding: covalent

- 16 The ions W^{2-} , X^{-} , Y^{+} and Z^{2+} are isoelectronic (they have the same number of electrons). Which option correctly lists the **elements** W, X, Y and Z in order of increasing atomic number, and the **ions** in order of increasing ionic radius?
- A Atomic number: $W < X < Y < Z$; Ionic radius: $Z^{2+} < Y^{+} < X^{-} < W^{2-}$
- B Atomic number: $W < X < Y < Z$; Ionic radius: $W^{2-} < X^{-} < Y^{+} < Z^{2+}$
- C Atomic number: $Z < Y < X < W$; Ionic radius: $Z^{2+} < Y^{+} < X^{-} < W^{2-}$
- D Atomic number: $Z < Y < X < W$; Ionic radius: $W^{2-} < X^{-} < Y^{+} < Z^{2+}$
- 17 A 0.010 mol sample of an ionic solid is dissolved in water to form a solution. The solution contains a total of 0.050 mol of ions. Which compound could be the ionic solid?
- A NaCl
- B MgCl_2
- C AlCl_3
- D Na_3PO_4
- E $\text{Al}_2(\text{SO}_4)_3$
- F $\text{C}_6\text{H}_{12}\text{O}_6$
- 18 An organic compound $\text{C}_x\text{H}_y\text{S}_z$ contains only carbon, hydrogen, and sulfur ($\text{C}_x\text{H}_y\text{S}_z$). Three separate oxidation processes are carried out using 1.0 mol of compound $\text{C}_x\text{H}_y\text{S}_z$ in each case:
- Process 1: Complete combustion of 1.0 mol of $\text{C}_x\text{H}_y\text{S}_z$ produces $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$, and $\text{SO}_2(\text{g})$ only, requiring exactly 7.0 mol of $\text{O}_2(\text{g})$.
 - Process 2: Incomplete combustion of 1.0 mol of $\text{C}_x\text{H}_y\text{S}_z$ produces $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$, and solid sulfur, $\text{S}(\text{s})$, only, requiring exactly 6.0 mol of $\text{O}_2(\text{g})$.
 - Process 3: Incomplete combustion of 1.0 mol of $\text{C}_x\text{H}_y\text{S}_z$ produces $\text{CO}(\text{g})$, $\text{H}_2\text{O}(\text{l})$, and solid sulfur, $\text{S}(\text{s})$, only, requiring exactly 4.0 mol of $\text{O}_2(\text{g})$.
- What is the molecular formula of compound $\text{C}_x\text{H}_y\text{S}_z$?
- A $\text{C}_3\text{H}_6\text{S}_2$
- B $\text{C}_3\text{H}_8\text{S}$
- C $\text{C}_4\text{H}_6\text{S}$
- D $\text{C}_4\text{H}_8\text{S}$
- E $\text{C}_4\text{H}_8\text{S}_2$
- F $\text{C}_4\text{H}_{10}\text{S}$
- G $\text{C}_5\text{H}_8\text{S}$
- H $\text{C}_5\text{H}_{10}\text{S}$

- 19 A researcher intends to purify organelle **Q** from a crude cellular lysate containing three major organelles: P, Q, and R. The physical properties of these organelles are shown in the table below.

Organelle	Mean Diameter (μm)	Density (g cm^{-3})
P	12.0	1.30
Q	2.5	1.15
R	0.05	1.50

The lysate is prepared in a buffer with a density of 1.05 g cm^{-3} .

Which of the following procedures represents the most effective **initial step** to purify organelle Q?

- A Centrifuge at $1\,000g$ (pellets particles $> 10\mu\text{m}$) and retain the supernatant.
- B Centrifuge at $10\,000g$ (pellets particles $> 2.0\mu\text{m}$) and retain the pellet.
- C Centrifuge at $100\,000g$ (pellets particles $> 0.02\mu\text{m}$) and retain the pellet.
- D Centrifuge at $1\,000g$ (pellets particles $> 10\mu\text{m}$) and retain the pellet.

- 20 An oxyanion has the formula XO_4^{3-} , where element X has atomic number Z and mass number A . All oxygen atoms are the isotope $^{16}_8\text{O}$.

The XO_4^{3-} ion contains a total of E electrons and a total of N neutrons.

Which row in the table gives the correct expressions for the atomic number Z , the number of neutrons in one atom of X, and the mass number A of element X?

	atomic number of X (Z)	neutrons in one atom of X	mass number of X (A)
A	$E-29$	$N-32$	$N+E-61$
B	$E-32$	$N-32$	$N+E-64$
C	$E-35$	$N-32$	$N+E-67$
D	$E-35$	$N-32$	$N+E-64$
E	$E-35$	$N-35$	$N+E-70$
F	$E-29$	$N-29$	$N+E-58$

21 Three separate organic reactions are carried out:

Reaction 1: but-2-ene is reacted with steam in the presence of an acid catalyst.

Reaction 2: butan-2-ol is heated under reflux with acidified potassium dichromate(VI).

Reaction 3: propanoic acid is heated with ethanol in the presence of a concentrated sulfuric acid catalyst.

Which of the following statements is/are correct?

1. The organic product of Reaction 1 contains a chiral carbon atom.
2. The organic product of Reaction 2 forms a silver mirror when warmed with Tollens' reagent.
3. The organic product of Reaction 3 is a structural isomer of pentanoic acid.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

22 A student mixes 40 cm^3 of 0.75 mol dm^{-3} NaOH solution with 60 cm^3 of 1.00 mol dm^{-3} HCl solution. Both solutions are initially at 20.0°C .

The enthalpy change of neutralisation for the reaction is -56 kJ mol^{-1} .

Assume that the density of the final mixture is 1.0 g cm^{-3} and the specific heat capacity is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

What is the final temperature of the mixture?

- A** 4.0°C
B 24.0°C
C 28.0°C
D 30.0°C
E 32.0°C

- 23** A 10 cm^3 sample of a gaseous hydrocarbon was completely burned in 80 cm^3 of oxygen gas (an excess) in a closed vessel.

The resulting mixture was cooled to the original room temperature and pressure, causing water vapour to condense completely. The remaining gas was then passed through an excess of aqueous potassium hydroxide, which absorbed all the carbon dioxide.

The volumes of gas measured at each stage, all under the same conditions of temperature and pressure, are shown in the table below:

Stage of experiment	Volume of gas / cm^3
Hydrocarbon vapour	10
Oxygen gas added initially (excess)	80
Gas mixture after combustion and cooling to room temperature	60
Gas mixture after treatment with excess aqueous potassium hydroxide	20

What is the molecular formula of the hydrocarbon?

- A** C_3H_6
B C_3H_8
C C_4H_6
D C_4H_8
E C_4H_{10}

- 24** An aqueous solution of an acid X forms a white precipitate when aqueous barium chloride, followed by dilute hydrochloric acid, is added to it.

Acid X reacts with solid substance Y to form exactly two products. One of these products forms a green precipitate when aqueous sodium hydroxide is added to it.

Which of the following could be X and Y?

	X	Y
A	HCl	CuO
B	HCl	Fe
C	HCl	FeCO_3
D	HCl	Zn
E	H_2SO_4	CuO
F	H_2SO_4	Fe
G	H_2SO_4	FeCO_3
H	H_2SO_4	Zn

- 25** Two monoprotic acids, X and Y , each have concentration $0.010 \text{ mol dm}^{-3}$ at the same temperature. The pH of acid X is 2 and the pH of acid Y is 3.

Which conclusion is best supported by this information?

- A** Acid X produces ten times the hydrogen-ion concentration of acid Y and is the stronger acid.
- B** Acid Y produces ten times the hydrogen-ion concentration of acid X and is the stronger acid.
- C** The acids have the same strength because their molar concentrations are equal.
- D** Acid X produces twice the hydrogen-ion concentration of acid Y .
- E** Acid Y does not dissociate in water.

- 26** A reaction between metallic zinc and dilute nitric acid produces an aqueous solution of zinc nitrate, water, and a gaseous oxide of nitrogen, X , as the only products.

In the balanced chemical equation for this reaction, 4 mol of zinc reacts to produce 5 mol of water.

What is the formula of the gaseous oxide of nitrogen X ?

- A** N_2O
- B** NO
- C** NO_2
- D** N_2O_3
- E** N_2O_5

- 27** Which of the following aqueous solutions has a pH of 1.0?

- A** 0.01 mol dm^{-3} hydrochloric acid
- B** 0.1 mol dm^{-3} hydrochloric acid
- C** 1.0 mol dm^{-3} hydrochloric acid
- D** 0.1 mol dm^{-3} ethanoic acid
- E** 0.1 mol dm^{-3} sodium hydroxide